CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 7: Longest Common Subsequence LCS Implementation

Course: Design Analysis and Algorithms Laboratory [23CS3505]
Category: Experiments / Practicals | Experiment #7 | Academic Year: 2026-27

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1. Overview & Objective:

Determine sequence similarity between strings using 2D dynamic programming grid.

2. Practical Tasks & Verification Steps:

1. Build m+1 ˆ n+1 integer table computing LCS length for two input strings X and Y.
2. Populate table matching characters diagonally 1 + dp[i-1][j-1] and mismatched cells from max dp[i-1][j], dp[i][j-1] .
3. Trace back through table reconstructing the longest common subsequence string.
4. Test algorithm on DNA sequences e.g. AGGTAB vs GXTXAYB verifying correct LCS output.

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